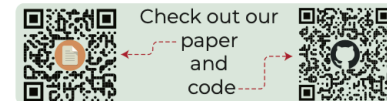
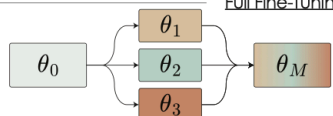


Accurate and Efficient Low-Rank Model Merging in Core Space

Aniello Panariello*, Daniel Marczak*, Simone Magistri, Angelo Porrello,
Bartłomiej Twardowski, Andrew D. Bagdanov, Simone Calderara, Joost van de Weijer

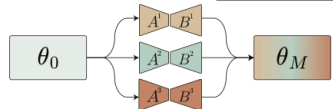


Full Fine-Tuning Merging



- ✗ Fine-Tuning Efficiency
- ✗ Merging Efficiency
- ✓ Merging Efficacy

LoRA Fine-Tuning Merging

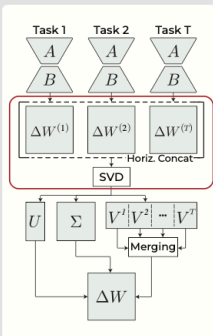


- ✓ Fine-Tuning Efficiency
- ✗ Merging Efficiency
- ✗ Merging Efficacy

KnOTS Merging

A recent step toward addressing the challenges in LoRA fine-tuning merging is KnOTS (Stoica et al., ICLR 2025), which aligns the task subspaces using SVD.

However, KnOTS must **operate on full-weight matrices**, thus it is **computationally heavy and scales poorly with model size**.



- ✓ Fine-Tuning Efficiency
- ✗✗ Merging Efficiency
- Merging Efficacy

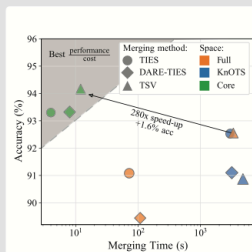
Our Core Space tackles all the challenges of merging LoRA Modules

Efficient: up to 600x speed improvement.

Accurate: consistent performance boost across vision and language benchmarks.

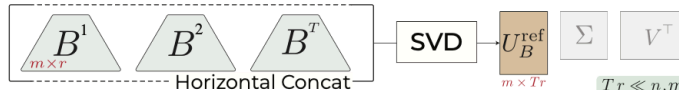
Universal: works with any merging technique (e.g., TIES, TSV, ISO, DARE).

In the Core Space we have a **denser representation with no loss of information** when projecting to it.

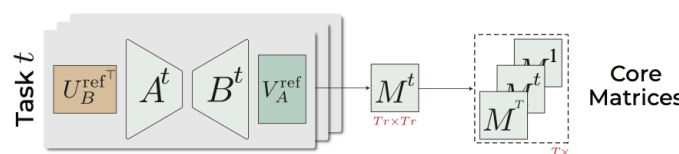


Core Space Goal: merging LoRA fine-tuned models **efficiently** and **accurately** while remaining in low-rank space.

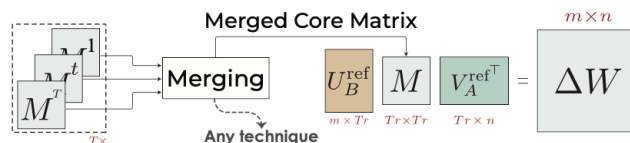
1 Compute reference bases



2 Project into Core Space



3 Merge & Reconstruct



Llama 3 8B

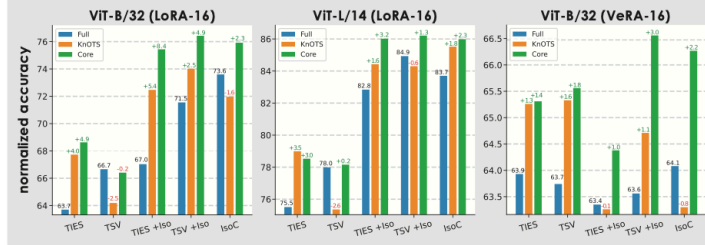
Method	Space	Avg (Δ Acc)	Time [s]	Rel. Time
TA	Full	90.38 (+0.00)	9	-
TIES	Full	91.08 (+0.00)	72	9
	KnOTS	92.53 (+1.45)	3000	375
	Core	93.30 (+2.22)	8	1
DARE-TIES	Full	89.44 (+0.00)	108	13
	KnOTS	91.10 (+1.66)	3180	397
	Core	93.32 (+3.88)	8	1
TSV	Full	92.55 (+0.00)	3360	280
	KnOTS	90.87 (-1.68)	4800	400
	Core	94.16 (+1.61)	12	1
Iso-C	Full	57.08 (+0.00)	540	67
	KnOTS	72.53 (+15.45)	4860	607
	Core	90.33 (+33.25)	8	1

• Reliable both in **Language** and **Vision** tasks.

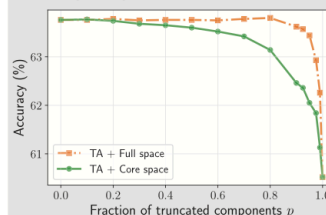
• Up to **600x faster** for Llama 3 8B.

• Works reliably across **VIT-B/32**, **VIT-L/14**.

• Handles **VeRA** and **different ranks**.



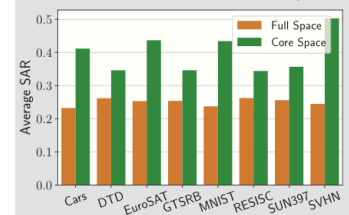
Core Space preserves task-relevant info



Core Space is **compact and dense**, truncating any component reduces accuracy, unlike full space, which contains many redundant directions.

This compactness facilitates model merging as it **extracts only the relevant components**.

Core Space improves subspace alignment



Core Space enforces a shared basis, resulting in **higher alignment** and **lower task interference**.

Thus, task subspaces overlap more effectively, which **reduces interference during merging** and leads to **better generalization**.

ideas.

micc

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